

Abhishek Kumar Upadhyay

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SUMMARY

Full-stack developer and self-starter who architects, builds, and ships scalable production systems end-to-end – SaaS, microservices, and payments/RBAC applications. Quick to pick up unfamiliar stacks, collaborates closely with cross-functional teams, and uses API testing tools and prompt engineering to accelerate delivery of reliable features.

EDUCATION

Indian Institute of Information Technology (IIIT), Bhopal

Aug 2023 – Jun 2027

Bachelor of Technology in Computer Science; CGPA: 8.27/10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering

PROJECTS

QuickAI – Full-Stack AI SaaS Platform | *React.js, Node.js, PostgreSQL, LLM APIs*

[GitHub](#) | [Live Demo](#)

- Architected and shipped a multi-tenant SaaS platform on the PERN stack, integrating LLM-based APIs to power AI features including article generation, image generation, background/object removal, and resume review.
- Designed, built, and tested 10+ REST API endpoints with role-based access control (RBAC), subscription-based access, and Clerk authentication, validating behavior with Postman before deployment.
- Deployed frontend on Vercel and backend on Render with environment-based configuration, delivering a stable, production-ready app used by real end users.

Cloud IDE – Online Code Execution Platform | *React, Node.js, Docker, Microservices*

[GitHub](#)

- Architected a cloud-based IDE supporting 4 languages (C, C++, JavaScript, Python) via 4 microservices (API Gateway, Execution, Project, Auth), keeping the platform stable and scalable while sandboxing untrusted code with Docker and Node.js child processes.
- Persisted application state across PostgreSQL, MongoDB, and Redis; used Redis to cache execution results, cutting repeat-run response time by approximately 30%.
- Built the editor UI with React.js and Monaco Editor, supporting multi-file projects up to 50MB per workspace, and verified execution correctness across all four language runtimes.

CourseHaven – E-Learning Marketplace | *MERN Stack*

[GitHub](#) | [Live Demo](#)

- Built a full-stack e-learning marketplace supporting 2 user roles (instructor, student) and 50+ seeded courses stored in MongoDB.
- Implemented instructor dashboards for CRUD operations and integrated the Stripe API for payment checkout and confirmation, testing edge cases across the payment flow.
- Wrote custom JWT-based authentication and RBAC middleware to gate instructor-only and student-only routes, verifying access control across both roles.

TECHNICAL SKILLS

Languages: C++, JavaScript (ES6+), Python, C, SQL, HTML5, CSS3

Engineering Practices: System Design Fundamentals, Scalable Architecture, API Testing, Debugging, Code Review, CI/CD Basics

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, EJS, Context API, Responsive Design

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, MVC Architecture, Server Actions, Microservices, API Gateway

Databases: MongoDB, PostgreSQL, Redis, Mongoose ODM, Prisma ORM

Tools & Platforms: Git, GitHub, Docker, Postman, Linux/Bash, VS Code, Vercel, Render

AI-Assisted Development: Prompt Engineering, LLM API Integration, Cursor, GitHub Copilot

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

LEADERSHIP & ACHIEVEMENTS

CODAME Coding Club, IIIT Bhopal

Sep 2024 – Oct 2025

Design Executive

- Collaborated cross-functionally to lead social media and design efforts for 5 coding workshops, coordinating outreach that contributed to a 25% increase in attendance.
 - Redesigned the UI of the club website and produced 20+ visual assets, improving on-page engagement and reducing bounce rate.
- Solved 900+ Data Structures and Algorithms problems across [LeetCode](#) and [GeeksforGeeks](#), spanning arrays, trees, graphs, and dynamic programming.